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2.11

a) $\hat{\mathcal{H}} = -\vec{\hat{\mu}} \cdot \vec{B}$, $\vec{\hat{\mu}} = -\frac{g}{\hbar}\left(\frac{|e|\hbar}{2m}\right)\vec{\hat{S}}$

$\vec{B} = B_y \hat{e}_y$

$\Rightarrow \hat{\mathcal{H}} = \frac{2\mu}{\hbar} \hat{S}_y \cdot B_y \hat{e}_y = \mu_B B_y \underbrace{\begin{pmatrix} 0 & -i \\ +i & 0 \end{pmatrix}}_{\hat{\sigma}_y} \hat{e}_y$

vp.1

$$\begin{vmatrix} -\lambda & -\mu_B B_y i \\ \mu_B B_y i & -\lambda \end{vmatrix} = \lambda^2 - \mu_B B_y = 0$$

$\Rightarrow \lambda_{\pm} = \pm \mu_B B_y$

$$\hat{U}(t-t_0) = e^{\frac{-i\hat{\mathcal{H}}(t-t_0)}{\hbar}} = e^{\frac{-i\mu_B B_y \hat{e}_y \cdot \hat{\sigma}_y \tau}{\hbar}}$$

on obtient que

$$U(t-t_0) = e^{-i\frac{\mu_B B_y}{\hbar}\tau \hat{e}_y \cdot \hat{\sigma}_y} = \left[\mathbb{1}\cos\left(\frac{\mu_B B_y}{\hbar}\tau\right) - i\sin\left(\frac{\mu_B B}{\hbar}\tau\right)\right]\left[\begin{matrix} 0 & -i \\ i & 0 \end{matrix}\right]$$

$$= \begin{bmatrix} \cos\left(\frac{\mu_B B_y}{\hbar}\tau\right) & -\sin\left(\frac{\mu_B B_y}{\hbar}\tau\right) \\ \sin\left(\frac{\mu_B B_y}{\hbar}\tau\right) & \cos\left(\frac{\mu_B B_y}{\hbar}\tau\right) \end{bmatrix} \rightarrow \text{rotation}$$

Rotation $\frac{\pi}{2}$:

$$\frac{\mu_B B_y}{\hbar}\tau = \frac{\pi}{2}$$

$$\Rightarrow \boxed{\tau = \frac{\hbar\pi}{2\mu_B B_y}}$$

b)

$$B_y = B_0 \cos\omega t$$

$$\hat{\mathcal{H}}(t) = \mu_B B_0 \cos\omega t\, \hat{\sigma}_y \cdot \hat{e}_y$$

$$|\psi(t)\rangle = \begin{bmatrix} a(t) \\ b(t) \end{bmatrix}, \quad |\psi(0)\rangle = \begin{bmatrix} a(0) \\ b(0) \end{bmatrix} = |z+\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} a(0) = 1 \\ b(0) = 0 \end{bmatrix}$$

$$i\hbar \frac{d}{dt}|\psi(t)\rangle = \hat{\mathcal{H}}(t)|\psi(t)\rangle$$

$$\Rightarrow i\hbar \frac{d}{dt}\begin{bmatrix} a(t) \\ b(t) \end{bmatrix} = \mu_B B_0 \cos\omega t \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}\begin{bmatrix} a(t) \\ b(t) \end{bmatrix}$$

$$i\hbar \begin{bmatrix} a'(t) \\ b'(t) \end{bmatrix} = \mu_B B_0 \cos\omega t \begin{bmatrix} -b(t) \\ a(t) \end{bmatrix}$$

$$\Rightarrow \begin{cases} a'(t) = \frac{\mu_B B_0 \cos\omega t}{i\hbar}(-b(t)) \\ b'(t) = \frac{\mu_B B_0 \cos\omega t}{i\hbar} a(t) \end{cases} \Rightarrow \begin{aligned} a(t) &= \cos\left[\frac{\mu_B B_0 \sin\omega t}{i\hbar}\right] \\ b(t) &= \sin\left[\frac{\mu_B B_0 \sin\omega t}{i\hbar}\right] \end{aligned}$$

↑ dérivé de l'argument

cyclique → sin, cos

$$|\psi(t)\rangle = \begin{bmatrix} \cos\left(\frac{\mu_B B_0 \sin\omega t}{i\hbar}\right) \\ \sin\left(\frac{\mu_B B_0 \sin\omega t}{i\hbar}\right) \end{bmatrix}$$

c)

état initial : $|z+\rangle$ → état stationnaire $|z+\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

2.2

$$\hat{J}_\pm |j,m\rangle = C_\pm |j, m\pm 1\rangle$$

$$\hat{J}_\pm = J_x \pm i J_y$$

$$\hat{J}_\pm \hat{J}_\mp = (\hat{J}_x \pm i\hat{J}_y)(\hat{J}_x \mp i\hat{J}_y) = \hat{J}_x^2 \mp i\hat{J}_x\hat{J}_y \pm i\hat{J}_y\hat{J}_x + \hat{J}_y^2$$

$$= \underbrace{\hat{J}_x^2 + \hat{J}_y^2}_{\hat{J}^2 - \hat{J}_z^2} \pm i\underbrace{[\hat{J}_y, \hat{J}_x]}_{-i\hbar J_z}$$

$$= \hat{J}^2 - J_z^2 \pm \hbar J_z$$

pour calculer $|a|^2 = \langle j,m|\hat{A}|j,m\rangle$

il faut faire +1 ($|j,m+1\rangle$)
puis -1 ($|j,m+1-1\rangle = |j,m\rangle$)
et vice versa

$$|C_\pm|^2 = \langle j,m | J_\mp J_\pm | j,m \rangle$$

$$= \langle j,m | \hat{J}^2 - \hat{J}_z^2 \pm \hbar \hat{J}_z | j,m \rangle$$

$$= \langle j,m | \hat{J}^2 | j,m \rangle - \langle j,m | \hat{J}_z^2 | j,m \rangle \pm \hbar \langle j,m | \hat{J}_z | j,m \rangle$$

$$= \langle j,m | \hbar^2 j(j+1) | j,m \rangle - \langle j,m | \hbar^2 m^2 | j,m \rangle \pm \hbar \langle j,m | \hbar m | j,m \rangle$$

$$= \hbar^2 j(j+1) - \hbar^2 m^2 \pm \hbar^2 m$$

$$|C_\pm|^2 = \hbar^2 \left(j(j+1) - m(m\pm 1) \right)$$

$$C_\pm = \sqrt{\hbar^2 \left(j(j+1) - m(m\pm 1) \right)}$$

$$\boxed{C_\pm = \hbar \sqrt{j(j+1) - m(m\pm 1)}}$$

2.3

$$\hat{\mathcal{H}} = \frac{2E_1}{\hbar^2} \hat{S}^{(1)} \cdot \hat{S}^{(2)} + \omega_0 \left(\hat{S}_z^{(1)} - \hat{S}_z^{(2)} \right)$$

a) Dans la base $\{|1,1\rangle, |1,0\rangle, |0,0\rangle, |1,-1\rangle\}$

$$\left(\hat{S}_z^{(1)} - \hat{S}_z^{(2)} \right) |1,1\rangle = \left(\hat{S}_z^{(1)} - \hat{S}_z^{(2)} \right) |++\rangle$$

$$= \frac{\hbar}{2} - \frac{\hbar}{2} |++\rangle$$

$$\left(\hat{S}_z^{(1)} - \hat{S}_z^{(2)} \right) |1,-1\rangle = \left(\hat{S}_z^{(1)} - \hat{S}_z^{(2)} \right) |--\rangle = -\frac{\hbar}{2} - \left(-\frac{\hbar}{2} \right) |--\rangle = 0$$

$$(\hat{S}_z^{(1)} - \hat{S}_z^{(2)})\,|1,0\rangle = (\hat{S}_z^{(1)} - \hat{S}_z^{(2)})\,\frac{1}{\sqrt{2}}\left(|+-\rangle + |-+\rangle\right)$$

$$= \frac{1}{\sqrt{2}}\left(\hat{S}_z^{(1)}|+-\rangle - \hat{S}_z^{(2)}|+-\rangle + \hat{S}_z^{(1)}|-+\rangle - \hat{S}_z^{(2)}|-+\rangle\right)$$

$$= \frac{1}{\sqrt{2}}\left(\frac{\hbar}{2}|+-\rangle - \left(-\frac{\hbar}{2}\right)|+-\rangle + \left(-\frac{\hbar}{2}\right)|-+\rangle - \frac{\hbar}{2}|-+\rangle\right)$$

$$= \frac{1}{\sqrt{2}}\left(\hbar|+-\rangle - \hbar|-+\rangle\right)$$

$$= \hbar\,|0,0\rangle$$

Même principe pour $(\hat{S}_z^{(1)} - \hat{S}_z^{(2)})|0,0\rangle = \hbar|1,0\rangle$, $\hat{S}_z^{(1)} - \hat{S}_z^{(2)} = \hbar\begin{pmatrix} 0&0&0&0\\ 0&0&1&0\\ 0&1&0&0\\ 0&0&0&0\end{pmatrix}$

$$\vec{S}_{(1)} \cdot \vec{S}_{(2)} = \frac{\hat{S}^2 - \hat{S}_{(1)}^2 - \hat{S}_{(2)}^2}{2}$$

La Base propre fait que la matrice $\vec{S}_{(1)} \cdot \vec{S}_{(2)}$ est diag

$$\frac{\hat{S}^2 - \hat{S}_{(1)}^2 - \hat{S}_{(2)}^2}{2}\,|1,1\rangle = \frac{1}{2}\left(2\hbar^2 - \frac{3}{4}\hbar^2 - \frac{3}{4}\hbar^2\right)|1,1\rangle$$

triplet

$$= \frac{\hbar^2}{4}\,|1,1\rangle$$

singulet

$$\frac{\hat{S}^2 - \hat{S}_{(1)}^2 - \hat{S}_{(2)}^2}{2}\,|0,0\rangle = \frac{1}{2}\left(0 - \frac{3}{4}\hbar^2 - \frac{3}{4}\hbar^2\right)|0,0\rangle$$

$$= -\frac{3\hbar^2}{4}\,|0,0\rangle$$

donc $\vec{S}_{(1)} \cdot \vec{S}_{(2)} = \hbar^2\begin{pmatrix} 1/4&0&0&0\\ 0&1/4&0&0\\ 0&0&-3/4&0\\ 0&0&0&1/4\end{pmatrix}$

donc $\hat{\mathcal{H}} = \frac{2E_1}{\hbar^2}\vec{S}^{(1)}\cdot\vec{S}^{(2)} + \omega_0(\hat{S}_z^{(1)} - \hat{S}_z^{(2)})$

$$= \frac{2E_1}{\hbar^2}\hbar^2\begin{pmatrix} 1/4 & 0 & 0 & 0 \\ 0 & 1/4 & 0 & 0 \\ 0 & 0 & -1/4 & 0 \\ 0 & 0 & 0 & 1/4 \end{pmatrix} + \omega_0\hbar\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\hat{\mathcal{H}} = \begin{pmatrix} E_1/2 & 0 & 0 & 0 \\ 0 & E_1/2 & \hbar\omega_0 & 0 \\ 0 & \hbar\omega_0 & -\frac{3E_1}{2} & 0 \\ 0 & 0 & 0 & E_1/2 \end{pmatrix}$$ Dans la base propre

b) vp.

$E_a = E_1/2$
$E_d = E_1/2$

Il reste 2 vp.

$$\begin{vmatrix} E_1/2 - \lambda & \hbar\omega_0 \\ \hbar\omega_0 & -3E_1/2 - \lambda \end{vmatrix} = \left(\frac{E_1}{2} - \lambda\right)\left(\frac{-3E_1}{2} - \lambda\right) - (\hbar\omega_0)^2 = 0$$

$$-\frac{3E_1^2}{4} - \frac{\lambda E_1}{2} + \frac{3\lambda E_1}{2} + \lambda^2 - (\hbar\omega_0)^2 = 0$$

$$\Rightarrow \lambda^2 + E_1\lambda + \left((\hbar\omega_0)^2 + \frac{3}{4}E_1^2\right)$$

$$E_{b,c} = \frac{-E_1 \pm \sqrt{E_1^2 + 4(\hbar\omega_0)^2 + 3E_1^2}}{2} = -\frac{E_1}{2} \pm \sqrt{E_1^2 + (\hbar\omega_0)^2}$$

c) État stationnaire

$E_a = 3/2\ \mu eV \longrightarrow$ état $|1,1\rangle$

$E_b = \left(-\frac{3}{2} + 5\right)\mu eV = \frac{7}{2}\mu eV$

$E_c = -\frac{13}{2}\mu eV$

$E_d = \frac{3}{2}\mu eV \longrightarrow$ état $|1,-1\rangle$

Pour E_b, E_c

$$\begin{bmatrix} \frac{3}{2} & 4 \\ 4 & -\frac{9}{2} \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \frac{7}{2} \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

$$\Rightarrow \frac{3}{2}\alpha + 4\beta = \frac{7}{2}\alpha$$

$$\Rightarrow 4\beta = 2\alpha$$

$$\Rightarrow \beta = \frac{\alpha}{2}$$

$$\longrightarrow \begin{bmatrix} \alpha \\ \frac{\alpha}{2} \end{bmatrix} = \alpha \begin{bmatrix} 1 \\ 1/2 \end{bmatrix} \Rightarrow |\alpha|^2 \begin{bmatrix} 1 & 1/2 \end{bmatrix} \begin{bmatrix} 1 \\ 1/2 \end{bmatrix} = |\alpha|^2 \frac{5}{4} = 1$$

$$\Rightarrow \alpha = \frac{2}{\sqrt{5}}$$

Pour E_b, état : $\frac{2}{\sqrt{5}}\left(|1,0\rangle + \frac{1}{2}|0,0\rangle\right)$

E_c :

$$\begin{bmatrix} 3/2 & 4 \\ 4 & -9/2 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = -\frac{13}{2} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} \Rightarrow \frac{3}{2}\alpha + 4\beta = -\frac{13}{2}\alpha$$

$$4\beta = -8\alpha$$

$$\beta = -2\alpha \Rightarrow \begin{bmatrix} \alpha \\ -2\alpha \end{bmatrix} = \alpha \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

Normalisation :

$$|\alpha|^2 \begin{bmatrix} 1 & -2 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \end{bmatrix} = |\alpha|^2 (5) \Rightarrow \alpha = \frac{1}{\sqrt{5}}$$

-1

l'état stationnaire est

$$\frac{1}{\sqrt{5}} \left(|1,0\rangle - 2|0,0\rangle \right)$$

d)

Dans la base composé :

$$\hat{S}_z^{(1)} = \hat{S}_z \otimes \mathbb{1} = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

$$\hat{S}_z^{(2)} = \mathbb{1} \otimes \hat{S}_z = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

pour que $S_z^{(1)} = -\frac{\hbar}{2}$ et $S_z^{(2)} = \frac{\hbar}{2}$, l'état propre correspondant est

$$\begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} = |-+\rangle$$

← base composé

Dans la base propre,

$$|1,0\rangle = \frac{1}{\sqrt{2}} \left(|+-\rangle + |-+\rangle \right)$$

$$|0,0\rangle = \frac{1}{\sqrt{2}} \left(|+-\rangle - |-+\rangle \right)$$

Donc, $|1,0\rangle - |0,0\rangle = \frac{1}{\sqrt{2}} \left(2|-+\rangle \right)$

$$\Rightarrow |-+\rangle = \frac{1}{\sqrt{2}} \left(|1,0\rangle - |0,0\rangle \right)$$

En résumé, l'état initial est

$B_{couplé} : |-+\rangle$

$B_{propre} : \frac{1}{\sqrt{2}}(|1,0\rangle - |0,0\rangle)$

c)

On écrit l'état initial dans la base des états stationnaires :

$$|E_b\rangle = \frac{2}{\sqrt{5}}\left(|1,0\rangle + \frac{1}{2}|0,0\rangle\right) = \frac{2}{\sqrt{5}}|1,0\rangle + \frac{1}{\sqrt{5}}|0,0\rangle$$

$$|E_c\rangle = \frac{1}{\sqrt{5}}(|1,0\rangle - 2|0,0\rangle) = \frac{1}{\sqrt{5}}|1,0\rangle - \frac{2}{\sqrt{5}}|0,0\rangle$$

$$\Rightarrow |0,0\rangle = \sqrt{5}\left(|E_b\rangle - \frac{2}{\sqrt{5}}|1,0\rangle\right)$$

$$= \sqrt{5}|E_b\rangle - 2|1,0\rangle$$

$$\Rightarrow |1,0\rangle = \sqrt{5}\left(|E_c\rangle + \frac{2}{\sqrt{5}}|0,0\rangle\right)$$

$$= \sqrt{5}|E_c\rangle + 2\left(\sqrt{5}|E_b\rangle - 2|1,0\rangle\right)$$

$$|1,0\rangle = \sqrt{5}|E_c\rangle + 2\sqrt{5}|E_b\rangle - 4|1,0\rangle$$

$$|1,0\rangle = \frac{1}{\sqrt{5}}|E_c\rangle + \frac{2}{\sqrt{5}}|E_b\rangle$$

$$\Rightarrow |0,0\rangle = \sqrt{5}|E_b\rangle - 2\left(\frac{1}{\sqrt{5}}|E_c\rangle + \frac{2}{\sqrt{5}}|E_b\rangle\right)$$

$$= \frac{1}{\sqrt{5}}|E_b\rangle - \frac{2}{\sqrt{5}}|E_c\rangle$$

L'état initial est donc,

$$|\psi(0)\rangle = \frac{1}{\sqrt{2}}\left(\frac{1}{\sqrt{5}}|E_c\rangle + \frac{2}{\sqrt{5}}|E_b\rangle - \frac{1}{\sqrt{5}}|E_b\rangle + \frac{2}{\sqrt{5}}|E_c\rangle\right)$$

$$- \frac{1}{\sqrt{2}}\left(\frac{1}{\sqrt{5}}|E_b\rangle + \frac{3}{\sqrt{5}}|E_c\rangle\right)$$

$$|\psi(t)\rangle = \hat{U}(t)|\psi(0)\rangle$$

$$= \exp\left(\frac{-i\hat{\mathcal{H}}t}{\hbar}\right)\left(\frac{1}{\sqrt{10}}|E_b\rangle + \frac{3}{\sqrt{10}}|E_c\rangle\right)$$

$$= \frac{1}{\sqrt{10}}\exp\left(\frac{-iE_b t}{\hbar}\right)|E_b\rangle + \frac{3}{\sqrt{10}}\exp\left(\frac{-iE_c t}{\hbar}\right)|E_c\rangle$$

$$P\left(S_z^{(1)} = \frac{\hbar}{2}\right) = |\langle +-|\psi(t)\rangle|^2 \quad \text{car} \quad \langle ++|\psi(t)\rangle = 0$$

$$|+-\rangle = \frac{1}{\sqrt{2}}\left(|1,0\rangle + |0,0\rangle\right)$$

$$= \frac{1}{\sqrt{10}}|E_c\rangle + \frac{2}{\sqrt{10}}|E_b\rangle + \frac{1}{\sqrt{10}}|E_b\rangle - \frac{2}{\sqrt{10}}|E_c\rangle$$

$$= \frac{3}{\sqrt{10}}|E_b\rangle - \frac{1}{\sqrt{10}}|E_c\rangle$$

$$|\langle +-|\psi(t)\rangle|^2 = \left|\frac{3}{10}\exp\left(\frac{-iE_b t}{\hbar}\right) - \frac{3}{10}\exp\left(\frac{-iE_c t}{\hbar}\right)\right|^2$$

$$= \frac{9}{100}\left|\exp\left(\frac{-iE_b t}{\hbar}\right) - \exp\left(\frac{-iE_c t}{\hbar}\right)\right|^2$$

$$= \frac{9}{100}\left|e^{-i(E_b - E_c)t/\hbar} - 1\right|^2$$

$$P(S_2^{(1)} = \frac{\hbar}{2}) = \frac{9}{25}\sin^2\left(\frac{(E_b - E_c)t}{2\hbar}\right) = \frac{9}{25}\sin^2\left(\frac{10t}{2\hbar}\right)$$

1) En effectuant le changement de base :

$$|\psi(t)\rangle = \frac{1}{\sqrt{10}} e^{-iE_b t/\hbar}\left(\frac{2}{\sqrt{5}}|1,0\rangle + \frac{1}{\sqrt{5}}|0,0\rangle\right) + \frac{3}{\sqrt{10}} e^{-iE_c t/\hbar}\left(\frac{1}{\sqrt{5}}|1,0\rangle - \frac{2}{5}|0,0\rangle\right)$$

$$= \frac{1}{5\sqrt{2}}\left(2e^{-iE_b t/\hbar} + 3e^{-iE_c t/\hbar}\right)|1,0\rangle + \frac{1}{5\sqrt{2}}\left(e^{-iE_b t/\hbar} - 6e^{-iE_c t/\hbar}\right)|0,0\rangle$$

base découplé : $\{|++\rangle, |+-\rangle, |-+\rangle, |--\rangle\}$

$$|\psi(t)\rangle = \frac{3}{10}\left(e^{-iE_b t/\hbar} - e^{-iE_c t/\hbar}\right)|+-\rangle + \frac{1}{10}\left(e^{-iE_b t/\hbar} + 9e^{-iE_c t/\hbar}\right)|-+\rangle$$

$$|x+\rangle = \frac{1}{\sqrt{2}}\left(|+\rangle + |-\rangle\right)$$

$$\left.\begin{aligned} |x+\rangle \otimes |+\rangle &= \frac{1}{\sqrt{2}}\left(|++\rangle + |-+\rangle\right) \\ |x+\rangle \otimes |-\rangle &= \frac{1}{\sqrt{2}}\left(|+-\rangle + |--\rangle\right) \end{aligned}\right\} \text{ permet } S_x^{(1)} = \frac{\hbar}{2}$$

$$P(S_x^{(1)} = \frac{\hbar}{2}) = \underbrace{\left|\left(\langle x+| \otimes \langle +|\right)|\psi(t)\rangle\right|^2}_{} + \underbrace{\left|\left(\langle x+| \otimes \langle -|\right)|\psi(t)\rangle\right|^2}_{}$$

$$= \left|\frac{1}{10\sqrt{2}}\left(e^{-iE_b t/\hbar} + 9e^{-iE_c t/\hbar}\right)\right|^2 + \left|\frac{3}{10\sqrt{2}}\left(e^{-iE_b t/\hbar} - e^{-iE_c t/\hbar}\right)\right|^2$$

$$= \frac{1}{200}\left(e^{-iE_b t/\hbar} + 9e^{-iE_c t/\hbar}\right)^2 + \frac{9}{200}\left(e^{-iE_b t/\hbar} - e^{-iE_c t/\hbar}\right)^2$$

2.4]

$$|\psi\rangle = \sqrt{\frac{1}{3}}\,|2,2,1\rangle \otimes |+\rangle + \sqrt{\frac{2}{3}}\,|2,2,0\rangle \otimes |-\rangle$$

$n=2, \ell=2$

a)

$$\hat{L}^2 |n,\ell,m\rangle = \ell(\ell+1)\hbar^2 |n,\ell,m\rangle$$

$$\hat{L}^2|\psi\rangle = \hat{L}^2\left(\sqrt{\frac{1}{3}}\,|2,2,1\rangle\otimes|+\rangle + \sqrt{\frac{2}{3}}\,|2,2,0\rangle\otimes|-\rangle\right)$$

$$= \sqrt{\frac{1}{3}}\,\hat{L}^2|2,2,1\rangle\otimes|+\rangle + \sqrt{\frac{2}{3}}\,\hat{L}^2|2,2,0\rangle\otimes|-\rangle$$

$$= \sqrt{\frac{1}{3}}\,2(2+1)\hbar^2|2,2,1\rangle\otimes|+\rangle + \sqrt{\frac{2}{3}}\,2(2+1)\hbar^2|2,2,0\rangle\otimes|-\rangle$$

$$\hat{L}^2|\psi\rangle = 6\hbar^2|\psi\rangle$$

une seule valeur de $\hat{L}^2$ possible

$$\mathbb{P}(\hat{L}^2 = 6\hbar^2) = 1$$

b) ← 2 valeurs

$$\hat{S}_z|\pm\rangle = \pm\frac{\hbar}{2}|\pm\rangle$$

$$\mathbb{P}(S_z = \tfrac{\hbar}{2}) = \sum_{m_\ell} |(\langle 2,2,m_\ell| \otimes \langle +1|)|\psi\rangle|^2$$

$$= |(\langle 2,2,1| \otimes \langle +1|)|\psi\rangle|^2$$

$$= \left|\sqrt{\frac{1}{3}}\right|^2 = \frac{1}{3}$$

$$P(\hat{S}_z = -\frac{\hbar}{2}) = \sum_{m_\ell} |(\langle 2,2,m_\ell | \otimes \langle -1|)|\psi\rangle|^2$$

$$= |(\langle 2,2,0| \otimes \langle -1|)|\psi\rangle|^2$$

$$= \left|\sqrt{\frac{2}{3}}\right|^2 = \frac{2}{3}$$

c) $\vec{J}^2$ ← utilisons la base propre $|J,M\rangle$

$$|\psi\rangle = \sqrt{\frac{1}{3}}|2,2,1\rangle|+\rangle + \sqrt{\frac{2}{3}}|2,2,0\rangle|-\rangle$$

$\ell = 2$, $s = \frac{1}{2}$ $\Rightarrow$ $J_{(1)} = 2$, $J_{(2)} = \frac{1}{2}$,

Pour

$$\sqrt{\frac{1}{3}}|2,2,1\rangle|+\rangle, \quad M_J^{(1)} = 1, \quad M_J^{(2)} = \frac{1}{2}$$

Pour $\sqrt{\frac{2}{3}}|2,2,0\rangle|-\rangle$, $M_J^{(1)} = 0$, $M_J^{(2)} = -\frac{1}{2}$

Avec le tableau de Clebsch-Gordan

$$|2,2,1\rangle|+\rangle = \sqrt{\frac{4}{5}}\left|\frac{5}{2}\,\frac{3}{2}\right) + \sqrt{\frac{1}{5}}\left|\frac{3}{2},\frac{3}{2}\right\rangle$$

$$|2,2,0\rangle|-\rangle = \sqrt{\frac{3}{5}}\left|\frac{5}{2},-\frac{1}{2}\right) + \sqrt{\frac{2}{5}}\left|\frac{3}{2},-\frac{1}{2}\right\rangle$$

Donc $|\psi\rangle = \sqrt{\frac{1}{3}}\left(\sqrt{\frac{4}{5}}\left|\frac{5}{2},\frac{3}{2}\right\rangle + \sqrt{\frac{1}{5}}\left|\frac{3}{2},\frac{3}{2}\right\rangle\right)$

$+ \sqrt{\frac{2}{3}}\left(\sqrt{\frac{3}{5}}\left|\frac{5}{2},-\frac{1}{2}\right\rangle + \sqrt{\frac{2}{5}}\left|\frac{3}{2},-\frac{1}{2}\right\rangle\right)$

$|\psi\rangle = \sqrt{\frac{4}{15}}\left|\frac{5}{2},\frac{3}{2}\right\rangle + \sqrt{\frac{1}{15}}\left|\frac{3}{2},\frac{3}{2}\right\rangle + \sqrt{\frac{6}{15}}\left|\frac{5}{2},\frac{1}{2}\right\rangle + \sqrt{\frac{4}{15}}\left|\frac{3}{2},-\frac{1}{2}\right\rangle$

$\hat{J}^2\left|\frac{5}{2},M\right\rangle = \frac{5}{2}\left(\frac{5}{2}+1\right)\hbar^2\left|\frac{5}{2},M\right\rangle = \frac{35}{4}\hbar^2$

$\hat{J}^2\left|\frac{3}{2},M\right\rangle = \frac{3}{2}\left(\frac{3}{2}+1\right)\hbar^2\left|\frac{3}{2},M\right\rangle = \frac{15}{4}\hbar^2$

$P\left(\hat{J}^2 = \frac{35}{4}\hbar^2\right) = \sum_M \left|\langle \tfrac{5}{2},M|\psi\rangle\right|^2$

$= \left|\langle \tfrac{5}{2},\tfrac{3}{2}|\psi\rangle\right|^2 + \left|\langle \tfrac{5}{2},-\tfrac{1}{2}|\psi\rangle\right|^2$

$= \left|\sqrt{\frac{4}{15}}\right|^2 + \left|\sqrt{\frac{6}{15}}\right|^2$

$= \frac{10}{15} = \frac{2}{3}$

$P\left(\hat{J}^2 = \frac{15}{4}\hbar^2\right) = \sum_M \left|\langle \tfrac{3}{2},M|\psi\rangle\right|^2$

$= \left|\langle \tfrac{3}{2},\tfrac{3}{2}|\psi\rangle\right|^2 + \left|\langle \tfrac{3}{2},-\tfrac{1}{2}|\psi\rangle\right|^2$

$= \frac{1}{15} + \frac{4}{15} = \frac{1}{3}$ ✓

d) $\hat{J}_z \left| J, \frac{3}{2} \right\rangle = \frac{3}{2}\hbar$

$\hat{J}_z \left| J, -\frac{1}{2} \right\rangle = -\frac{\hbar}{2}$

$$P\left(\hat{J}_z = \tfrac{3}{2}\hbar\right) = \sum_J \left| \left\langle J, \tfrac{3}{2} \middle| \psi \right\rangle \right|^2$$

$$= \left| \left\langle \tfrac{5}{2}, \tfrac{3}{2} \middle| \psi \right\rangle \right|^2 + \left| \left\langle \tfrac{3}{2}, \tfrac{3}{2} \middle| \psi \right\rangle \right|^2$$

$$= \frac{4}{15} + \frac{1}{15} = \frac{1}{3}$$

$$P\left(\hat{J}_z = -\tfrac{\hbar}{2}\right) = \sum_J \left| \left\langle J, -\tfrac{1}{2} \middle| \psi \right\rangle \right|^2$$

$$= \left| \left\langle \tfrac{5}{2}, -\tfrac{1}{2} \middle| \psi \right\rangle \right|^2 + \left| \left\langle \tfrac{3}{2}, -\tfrac{1}{2} \middle| \psi \right\rangle \right|^2$$

$$= \frac{6}{15} + \frac{4}{15} = \frac{2}{3}$$

2.5 |

$$\hat{\mathcal{H}} = \hbar\omega\left(\hat{a}_1^\dagger \hat{a}_1 + \frac{1}{2}\right) + \hbar\omega\left(\hat{a}_2^\dagger \hat{a}_2 + \frac{1}{2}\right)$$

a) · singulet antisymétrique

$$|\Psi\rangle = |\Psi_{\text{spatiale}}\rangle \otimes |\Psi_{\text{spin}}\rangle$$

↑ Symétrique

$$|\Psi_{\text{spatiale}}\rangle = |0\rangle_1 |0\rangle_2$$
$$|\Psi_{\text{spin}}\rangle = |00\rangle$$

$$|\Psi\rangle = |0\rangle_1 |0\rangle_2 \otimes |00\rangle$$

$E = ?$

$$\hat{\mathcal{H}}|\Psi\rangle = \hat{\mathcal{H}}\, |0\rangle_1 |0\rangle_2 \otimes |00\rangle$$

$$= \left(\frac{\hbar\omega}{2} + \frac{\hbar\omega}{2}\right) |0\rangle_1 |0\rangle_2 \otimes |00\rangle$$

$$\hat{\mathcal{H}}|\Psi\rangle = \hbar\omega\,|\Psi\rangle$$

↑ E

$\Psi(x) = ?$

$$\Psi(x) = \langle x|\Psi\rangle = \langle x|\left(|0\rangle_1 |0\rangle_2 \otimes |00\rangle\right)$$

$$= \Psi_0(x_1)\,\Psi_1(x_2)$$

$$= \left[\left(\frac{\alpha}{\pi}\right)^{1/4} e^{-\frac{\alpha x_1^2}{2}}\right]\left[\left(\frac{\alpha}{\pi}\right)^{1/4} e^{-\frac{\alpha x_2^2}{2}}\right] = \left(\frac{\alpha}{\pi}\right)^{1/2} e^{-\frac{\alpha(x_1^2+x_2^2)}{2}}$$

b) triplet → symétrique } antisimetrique (e^-)
=> spatial → antisymétrique }

$$|\Psi\rangle = |\Psi_{spatial}\rangle \otimes |\Psi_{spin}\rangle$$

$$\overset{①}{=} \frac{1}{\sqrt{2}}\left(|0\rangle_1|1\rangle_2 - |1\rangle_1|0\rangle_2\right) \otimes |11\rangle$$

$$\overset{②}{=} \frac{1}{\sqrt{2}}\left(|0\rangle_1|1\rangle_2 - |1\rangle_1|0\rangle_2\right) \otimes |1-1\rangle$$

$$\overset{③}{=} \frac{1}{\sqrt{2}}\left(|0\rangle_1|1\rangle_2 - |1\rangle_1|0\rangle_2\right) \otimes |10\rangle$$

$$\hat{\mathcal{H}}|\Psi\rangle = \left(\frac{\hbar\omega}{2} + \hbar\omega\left(1+\tfrac{1}{2}\right)\right)|\Psi\rangle = \underbrace{2\hbar\omega}_{E}|\Psi\rangle$$

$$\langle x|\Psi\rangle = \langle x|\left(\frac{1}{\sqrt{2}}\left[|0\rangle_1|1\rangle_2 - |1\rangle_1|0\rangle_2\right] \otimes \cdots\right)$$

$$= \frac{1}{\sqrt{2}}\left(\psi_0(x_1)\psi_1(x_2) - \psi_1(x_1)\psi_0(x_2)\right)$$

$$= \frac{1}{\sqrt{2}}\left(\left(\frac{\alpha}{\pi}\right)^{1/4} e^{-\frac{\alpha x_1^2}{2}} \left(\frac{\alpha}{\pi}\right)^{1/4} \sqrt{2\alpha}\, x_2 e^{-\frac{\alpha x_2^2}{2}} - \left(\frac{\alpha}{\pi}\right)^{1/4} e^{-\frac{\alpha x_2^2}{2}} \left(\frac{\alpha}{\pi}\right)^{1/4} \sqrt{2\alpha}\, x_1 e^{-\frac{\alpha x_1^2}{2}}\right)$$

$$\Psi(x_1,x_2) = \frac{\sqrt{2\alpha}}{\sqrt{2}}\left(\frac{\alpha}{\pi}\right)^{1/2}\left[x_2 e^{-\frac{\alpha}{2}(x_1^2+x_2^2)} - x_1 e^{-\frac{\alpha}{2}(x_2^2+x_1^2)}\right]$$

$$\Psi(x_1,x_2) = \sqrt{\alpha}\left(\frac{\alpha}{\pi}\right)^{1/2}(x_2-x_1)\, e^{-\frac{\alpha}{2}(x_1^2+x_2^2)}$$

c)

$$\mathcal{H}_{int} = k e^{-\beta(x_1 - x_2)^2}$$

$$|\Psi\rangle = |0\rangle_1 |0\rangle_2 \otimes |00\rangle$$

$$\langle \mathcal{H}_{int} \rangle = \langle \Psi | \mathcal{H}_{int} | \Psi \rangle$$

$$= \left({}_1\langle 0 |\, {}_2\langle 0 | \otimes \langle 00 | \right) \underbrace{\mathcal{H}_{int}}_{\text{ne dépend pas du } t} \left(|0\rangle_1 |0\rangle_2 \otimes |00\rangle \right)$$

$$= \langle \Psi | \int dx_1' dx_2' \, |x_1', x_2'\rangle \langle x_1', x_2'| \, \mathcal{H}_{int} \int dx_1 dx_2 \, |x_1, x_2\rangle \langle x_1, x_2 | \Psi \rangle$$

$$\langle x_1', x_2' | x_1, x_2 \rangle = \delta(x_1' - x_1)\, \delta(x_2' - x_2)$$

$$\langle \mathcal{H}_{int} \rangle = \int dx_1 dx_2 \; k e^{-\beta(x_1 - x_2)^2} \, |\Psi(x_1, x_2)|^2$$

$$= \int dx_1 dx_2 \; k e^{-\beta(x_1 - x_2)^2} \, |\Psi_0(x_1)|^2 \, |\Psi_0(x_2)|^2$$

$$= \int dx_1 dx_2 \; k e^{-\beta(x_1 - x_2)^2} \left(\frac{\alpha}{\pi}\right)^{1/2} e^{-\alpha x_1^2} \left(\frac{\alpha}{\pi}\right)^{1/2} e^{-\alpha x_2^2}$$

$$= \int dx_1 dx_2 \; \frac{k\alpha}{\pi} \, e^{-\beta(x_1 - x_2)^2} \, e^{-\alpha(x_1 + x_2)^2}$$

$$\langle \mathcal{H}_{int} \rangle = \int dx_1 dx_2 \; \frac{k\alpha}{\pi} \, e^{-(\beta + \alpha)(x_1 + x_2)^2}$$

2.6/

a) $E_0 = -3J$

Dégénérescence : 4

b)

$$|\psi_1\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}, |\psi_2\rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}, |\psi_3\rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \\ \alpha \\ 0 \\ \alpha \\ \alpha \\ 0 \end{pmatrix}, |\psi_4\rangle = \begin{pmatrix} 0 \\ \alpha \\ \alpha \\ 0 \\ \alpha \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$\alpha = -0,57735027$

```python
import numpy as np
import qutip as qt
import matplotlib.pyplot as plt

# construction de l'hamiltonien dépendant du temps H(t) = σy * cos(ωt)

def Htd(t, args):
    om = 1
    return qt.sigmay()*np.cos(om*t)

# etat initial |z+>

psi0 = qt.basis(2, 0)

t = np.linspace(0, 10, 500)

output = qt.sesolve(Htd, psi0, t)

# calcul de la probabilité

P = []

for state in output.states:
    b = state[1, 0]
    P.append(np.abs(b)**2)

plt.plot(t, P)
plt.xlabel('Temps')
plt.ylabel('Probabilité')
plt.title('Évolution de la probabilité')
plt.show()
```

```python
import qutip as qt
import numpy as np

# Matrice identité 2x2 et de Pauli
I = qt.qeye(2)
Sx = qt.sigmax()
Sy = qt.sigmay()
Sz = qt.sigmaz()

# produit tensoriel pour trois particules de spin 1/2

Sx1 = qt.tensor(Sx, I, I)
Sy1 = qt.tensor(Sy, I, I)
Sz1 = qt.tensor(Sz, I, I)

Sx2 = qt.tensor(I, Sx, I)
Sy2 = qt.tensor(I, Sy, I)
Sz2 = qt.tensor(I, Sz, I)

Sx3 = qt.tensor(I, I, Sx)
Sy3 = qt.tensor(I, I, Sy)
Sz3 = qt.tensor(I, I, Sz)

# calcul des multiplications des matrice de Pauli

S1_S2 = Sx1*Sx2 + Sy1*Sy2 + Sz1*Sz2
S1_S3 = Sx1*Sx3 + Sy1*Sy3 + Sz1*Sz3
S2_S3 = Sx2*Sx3 + Sy2*Sy3 + Sz2*Sz3

# construction de l'hamiltonien

H = -(S1_S2 + S1_S3 + S2_S3)

# calcul des valeurs propres de l'hamiltonien

valeurs_propres, vecteurs_propres = H.eigenstates()

# Energie du niveau fondamental et sa degenerescence

E0 = valeurs_propres[0]
degenerescence = np.sum(np.abs(valeurs_propres - E0) < 1e-10)

print(f"Énergie du fondamental (en unités de J) : {E0}")
print(f"Dégénérescence : {degenerescence}")
print("\nVecteurs propres du fondamental :")

# contenu des vecteurs propres du fondamental

for i in range(len(vecteurs_propres)):
    if np.abs(valeurs_propres[i] - E0) < 1e-10:
        print(vecteurs_propres[i])
```